GCD QUOTIENTS: A QUANTITATIVE LOWER BOUND AND SHARP FIXED-DIMENSIONAL ORDERS

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ABSTRACT. For a finite set A of positive integers, let $Q(A) = \{a/\gcd(a,b) : a,b \in A\}$, and let $h(n)$ be the minimum of $|Q(A)|$ over n-element sets. We prove

$$h(n) \geq \sqrt{n}\exp\left(c\sqrt{\frac{\log n}{\log\log n}}\right)$$

for all sufficiently large n and an absolute $c > 0$. The proof uses an elementary extraction onto logarithmically many original coordinates and an estimate for positive differences with controlled fiber irregularity. For integers whose prime divisors lie in any prescribed set of $d \geq 1$ distinct primes, we determine the exact order $h_d(n) = \Theta_d(n^{u_d})$, where $u_1 = 1$ and $u_d = \min_{p+s=d}(u_p + u_s)/(1 + u_p + u_s)$ over positive splits. Matching constructions and a two-corner pruning argument establish this for every fixed d. Both headline results, including an explicit unrestricted constant $c = 1/16000$, have been formalized in Lean. We also give dimension-sensitive bounds for structured sets and a completion-and-exchange criterion for further progress. The sharper unrestricted scale $\sqrt{n}\exp(\Theta(\sqrt{\log n}))$ remains conjectural.

1. INTRODUCTION

For a finite set $A \subset \mathbb{N}_{>0}$ write

$$Q(A) = \left\{\frac{a}{\gcd(a,b)} : a,b \in A\right\}, \qquad h(n) = \min_{|A|=n} |Q(A)|.$$

Erdős asked for the order of magnitude of $h(n)$ [Er73, Section 5, p. 125]; see also [EP]. Prime valuations turn this arithmetic question into a problem about coordinatewise positive differences. For $x \in \mathbb{R}^d$ put $x^+ = (\max(x_1, 0), \ldots, \max(x_d, 0))$, and define

$$\mathcal{D}(F) = (F - F)^+, \qquad M(F) = \max_{a\in F} |(F - a)^+|.$$

We call $(F - a)^+$ a forward fan. The equality $v_p(a/\gcd(a,b)) = (v_p(a) - v_p(b))^+$ identifies $h(n)$ with the minimum of $|\mathcal{D}(F)|$ over all n-point integer sets in arbitrary dimension. Restricting to positive integers whose prime divisors belong to a prescribed set of d distinct primes gives

$$h_d(n) = \min_{\substack{F\subset\mathbb{Z}^d \\ |F|=n}} |\mathcal{D}(F)|, \qquad h(n) = \min_{d\geq 1} h_d(n).$$

The geometric lower bounds below hold for real point sets as well.

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Context. The classical bounds are $\sqrt{n} \ll h(n) \ll n^{2/3}$; Granville and Roesler give proofs and the geometric formulation [GR99]. The sharp planar lower bound is due to Granville and Roesler; the matching construction is due to Freiman and Lev, as reported in [GR99]. Holzman, Lev and Pinchasi [HLP08] proved

$$h_3(n) \geq \tfrac{1}{6}n^{3/5}, \qquad h_4(n) \gg n^{6/11}(\log n)^{-2/11}.$$

The 2026 preprint of Schmitt et al. [PC26, Appendix A] establishes

$$h(n) \leq \sqrt{n}\exp(C\sqrt{\log n}), \tag{1}$$

and hence $h(n) = n^{1/2+o(1)}$. Their construction gives upper exponents $2/3, 4/7, 8/15, \ldots$ in dimensions $2, 5, 11, \ldots$ [PC26, Propositions A.7–A.8]; these are upper bounds, without matching lower bounds in each dimension. Our construction refines this layering method: it combines sets in dimensions p and s in dimension $p+s$, permits unequal pieces, and optimizes over all positive splits. The fixed-dimensional contribution is the resulting recursive exponent together with a matching lower bound for every d, including $5/8$ in dimension three and $4/7$ in dimension four. The planar exponent and the underlying layering idea have these predecessors.

Pham's unrefereed manuscript [Ph26] claims the lower bound $\sqrt{n}\exp(c(\log n)^{1/3})$, using a dimension-dependent estimate and weak polynomial Freiman–Ruzsa theory [GGMT25]; its main theorem is not yet formalized in the accompanying Lean project. Our unrestricted lower bound has the stronger gain $\exp(c\sqrt{\log n/\log\log n})$. Kitamura's separate Lean developments [Ki26, KiExp26] verify the logarithmic exponent $1/2$ and the divergence $h(n)/\sqrt{n} \to \infty$, with the appropriate comparison to the zero-inclusive Formal Conjectures definition. These conclusions do not determine the order of $h(n)$.

Schmitt et al. [PC26, Appendix A, "Relation to prior announcements"] document earlier unpublished joint work of Bollobás and Leader announcing an improvement to the $n^{2/3}$ upper bound; see also Leader's 2009 and 2012 seminar abstracts [Le09, Le12]. We know of no written account specifying its bound and make no priority claim over the unknown contents of that work. Our claims concern the quantitative lower bound and the sharp fixed-dimensional orders stated below, relative to the public results cited here.

Main results. Our first result improves the quantitative lower bound.

Theorem 1.1 (Unrestricted lower bound)**.** *There is an absolute $c > 0$ such that, for all sufficiently large n,*

$$h(n) \geq \sqrt{n}\exp\left(c\sqrt{\frac{\log n}{\log\log n}}\right).$$

The formal proof supplies $c = 1/16000$; we do not optimize constants. Combined with (1), this gives

$$\log\left(h(n)/\sqrt{n}\right) = (\log n)^{1/2+o(1)}.$$

It leaves a factor $\sqrt{\log\log n}$ between the lower and upper bounds on this logarithm.

For the fixed-dimensional problem, define

$$u_1 = 1, \qquad u_d = \min_{\substack{p+s=d\\ p,s\geq 1}} \frac{u_p + u_s}{1 + u_p + u_s} \quad (d \geq 2). \tag{2}$$

Theorem 1.2 (Sharp order in every fixed dimension)**.** *For every $d \geq 1$, there is a constant C_d such that $h_d(n) \leq C_d n^{u_d}$ for all $n \geq 1$. Every finite nonempty $F \subset \mathbb{R}^d$ satisfies*

$$|\mathcal{D}(F)| \geq M(F) \geq 2^{-2^{d+1}}|F|^{u_d}.$$

Consequently $h_d(n) = \Theta_d(n^{u_d})$ for every fixed d.

Here $u_d \geq d/(2d-1)$, with equality for powers of two. The first values illustrate the nonuniform recursion: In particular, $h_3(n) \asymp n^{5/8}$ and $h_4(n) \asymp n^{4/7}$. Thus the theorem determines a family

d	1	2	3	4	5	6	7	8	16
u_d	1	$\frac{2}{3}$	$\frac{5}{8}$	$\frac{4}{7}$	$\frac{31}{55}$	$\frac{26}{47}$	$\frac{67}{123}$	$\frac{8}{15}$	$\frac{16}{31}$

TABLE 1. Sharp fixed-dimensional exponents.

of extremal functions, one for each fixed prime support. The constants depend on d; taking a minimum over growing d does not settle the unrestricted problem. Appendix A gives the better lower constant 8^{-d} through dimension fifteen.

The estimate used for Theorem 1.1 has more favorable dimension dependence. For $I \subseteq [d]$, let $P_I(S) = |\pi_I S|$ and let $\Delta_I(S)$ be the largest fiber of π_I on S. Put

$$R(S) = \max_{I \subseteq [d]} \frac{P_I(S)\Delta_I(S)}{|S|}.$$

Disjoint translates of a largest fiber show that $1 \leq R(S) \leq |S - S|/|S|$.

Theorem 1.3 (Fiber-regularity estimate)**.** *Let $d \geq 1$, with $\alpha_1 = 0$ and $\alpha_d = (1 + \log_2 d)/2$ for $d \geq 2$. Every finite nonempty $S \subset \mathbb{R}^d$ satisfies*

$$M(S) \geq 256^{-d} R(S)^{-\alpha_d} |S|^{u_d}.$$

Corollary 1.4. *Let $d \geq 1$ and $B_d = 1 + 2\alpha_d$. Every finite nonempty $F \subset \mathbb{R}^d$ satisfies*

$$|\mathcal{D}(F)| \geq 256^{-d/B_d} |F|^{\frac{1}{2} + \frac{u_d - 1/2}{B_d}} \geq 256^{-d/B_d} |F|^{\frac{1}{2} + \frac{1}{(4d-2)B_d}}.$$

The exponent gain is of order $1/(d \log d)$. An elementary extraction theorem supplies an actual large subset on $O(\log K)$ original coordinates when $|S - S| \leq K|S|$. Combining the two estimates proves Theorem 1.1, without a PFR input. This insistence on original coordinates matters: general additive embeddings need not preserve positive parts.

Scope and organization. Sections 3–5 prove the sharp fixed-dimensional theorem. Sections 6 and 7 prove the unrestricted bound. Section 8 treats midpoint-closed sets and ordered layers. Section 9 develops critical cores and a concrete exchange criterion, isolating the missing statement needed for the conjectural scale

Conjecture 1.5. $h(n) = \sqrt{n} \exp(\Theta(\sqrt{\log n}))$.

Section 10 records the precise formalization boundary. The new completion-and-exchange arguments have written proofs here; they are distinguished from the results in the completed Lean audit.

2. Preliminaries

All sets are finite and nonempty. For $F \subset \mathbb{R}^d$ and $a \in F$ write $(F - a)^+ = \{(x - a)^+ : x \in F\}$ and $(a - F)^+ = \{(a - x)^+ : x \in F\}$. Besides $M(F)$ we use the two-sided fan

$$T(F) = \max_{a \in F} \max\left\{|(F - a)^+|,\ |(a - F)^+|\right\}.$$

For $I \subseteq [d]$ let π_I be the coordinate projection onto $\mathbb{R}^I$, $P_I(F) = |\pi_I F|$, and let $\Delta_I(F)$ be the largest size of a fiber $\{x \in F : \pi_I x = z\}$. For $I = \varnothing$ we put $P_\varnothing = 1$ and $\Delta_\varnothing = |F|$.

Lemma 2.1. *For finite $A \subset \mathbb{N}_{>0}$ let F_A be the set of prime-exponent vectors of its elements. Then $|F_A| = |A|$ and $|Q(A)| = |\mathcal{D}(F_A)|$. Conversely, every finite $F \subset \mathbb{Z}^d$ is a translate of F_A for some A supported on d given primes. Since $\mathcal{D}$ is translation invariant, $h_d(n) = \min\{|\mathcal{D}(F)| : F \subset \mathbb{Z}^d,\ |F| = n\}$ and $h(n) = \min_d h_d(n)$.*

Proof. Unique factorization and $v_p(a/\gcd(a,b)) = (v_p(a) - v_p(b))^+$. □

Lemma 2.2. *Let $F \subset \mathbb{R}^d$ with $|F| = n$.*

1. *$M(F) \le T(F) \le |\mathcal{D}(F)|$, and $\mathcal{D}(-F) = \mathcal{D}(F)$.*
2. *$|F - F| \le |\mathcal{D}(F)|^2$.*
3. *For every $a \in F$, $|(F-a)^+| \cdot |(a-F)^+| \ge n$; in particular $T(F) \ge n^{1/2}$.*
4. *For every coordinate i, $M(F) \ge P_{\{i\}}(F)$.*

Proof. (1) is clear. For (2) and (3) use $z = z^+ - (-z)^+$: the maps $z \mapsto (z^+, (-z)^+)$ on $F - F$ and $x \mapsto ((x-a)^+, (a-x)^+)$ on F are injective. For (4), anchor at a point a with minimal i-th coordinate; then $x \mapsto (x-a)_i^+ = x_i - a_i$ takes $P_{\{i\}}(F)$ values. □

We use the standard Plünnecke–Ruzsa consequence [Ru89, Pe12]

$$|A - A| \le K|A| \quad \Longrightarrow \quad |A + A| \le K^2|A| \tag{3}$$

for finite subsets of an abelian group. All logarithms are natural unless a base is specified.

3. Constructions

The following proposition generalizes the separated-suspension construction of [PC26, Lemma A.6]. Taking sufficiently separated levels makes the additional level coordinate unnecessary and allows the two factors to have different dimensions.

Proposition 3.1 (Gluing)**.** *Let $p, s \ge 1$, and let $F \subset \mathbb{Z}^p$ and $G \subset \mathbb{Z}^s$ be finite and nonempty, let L bound every coordinate diameter of F and of G, let $M > 2L$ and $K \ge 1$ be integers, and put*

$$T = \big\{(x + jM\mathbf{1}_p,\ y - jM\mathbf{1}_s) : x \in F,\ y \in G,\ 0 \le j < K\big\} \subset \mathbb{Z}^{p+s}.$$

Then $|T| = K|F||G|$, $|T - T| = (2K-1)|F-F||G-G|$, and

$$|\mathcal{D}(T)| = |\mathcal{D}(F)|\,|\mathcal{D}(G)| + (K-1)\big(|F-F| + |G-G|\big).$$

Proof. A difference of two points of T at levels j, j' equals $(u + gM\mathbf{1}_p,\ v - gM\mathbf{1}_s)$ with $u \in F - F$, $v \in G - G$, $g = j - j'$. Every coordinate of u and v lies in $[-L, L]$ and $M > 2L$, so the first coordinate determines g, and then u and v; this gives the formulas for $|T|$ and $|T - T|$. If $g = 0$ the positive part is (u^+, v^+), and all of $\mathcal{D}(F) \times \mathcal{D}(G)$ occurs. If $g > 0$ the first block is positive and the second negative, so the positive part is $(u + gM\mathbf{1}_p, 0)$; for each $g \in \{1, \dots, K-1\}$ all $u \in F - F$ occur. Symmetrically $g < 0$ gives $(0, v + |g|M\mathbf{1}_s)$. These classes are pairwise disjoint because $M > 2L$: in the class $g = 0$ every coordinate lies in $[0, L]$; for $g > 0$ the first block has all coordinates in $[gM - L, gM + L]$, which also determines g; and for $g < 0$ the first block vanishes while the second block has all coordinates in $[|g|M - L, |g|M + L]$. □

Lemma 3.2. *u_d is nonincreasing, and $d/(2d-1) \le u_d \le 1$, with equality in the lower bound if d is a power of two. In particular $u_d \ge \frac{1}{2} + \frac{1}{4d-2}$.*

Proof. The upper bound is immediate from the recursion. For monotonicity, $u_2 \le u_1$; given a minimizing split $p + s = d$, induction gives $u_{p+1} \le u_p$, and the split $(p+1) + s = d + 1$ gives $u_{d+1} \le u_d$. The map $t \mapsto t/(1+t)$ is increasing, so by induction it suffices to show $\frac{A}{1+A} \ge \frac{d}{2d-1}$ for $A = \frac{p}{2p-1} + \frac{s}{2s-1}$ and $d = p + s$, i.e. $A \ge 1 + \frac{1}{d-1}$. Now $A - 1 = \frac{1}{2(2p-1)} + \frac{1}{2(2s-1)} \ge$

$\frac{4}{2(2p-1)+2(2s-1)} = \frac{1}{d-1}$ by the harmonic–arithmetic mean inequality. For $d = 2m$ the balanced split gives $\frac{2u_m}{1+2u_m} = \frac{2m}{4m-1}$ when $u_m = \frac{m}{2m-1}$. □

Theorem 3.3. *For every $d \geq 1$ there is C_d with $h_d(n) \leq C_d n^{u_d}$ for all $n \geq 1$.*

Proof. We construct, for every binary tree τ with d leaves, positive integers a_τ, b_τ with $b_\tau/a_\tau = \alpha_\tau$, and sets $F_W \subset \mathbb{Z}^d$ $(W \geq 1)$ with

$$|F_W| = W^{a_\tau}, \qquad |\mathcal{D}(F_W)| \leq C_\tau W^{b_\tau}, \qquad |F_W - F_W| \leq C_\tau W^{a_\tau},$$

where $\alpha_{\text{leaf}} = 1$ and $\alpha_\tau = (\alpha_1 + \alpha_2)/(1 + \alpha_1 + \alpha_2)$ for a tree with subtrees τ_1, τ_2. Minimizing over trees gives exponent u_d, by (2) and induction.

For a leaf take $F_W = \{0, \dots, W-1\}$, with $(a, b) = (1, 1)$. Given families for τ_1, τ_2 with parameters (a_i, b_i), use the first family at parameter W^{a_2} and the second at parameter W^{a_1}; both then have W^A points, $A = a_1 a_2$, and difference-set exponents $b_1 a_2$ and $b_2 a_1$. Put $B = b_1 a_2 + b_2 a_1$. Since $\alpha_i \geq 1/2$ (inductively, as $\alpha_1 + \alpha_2 \geq 1$), $B \geq A$. Glue with $K = W^{B-A}$ levels (Proposition 3.1). The result has W^{A+B} points, $|\mathcal{D}| \leq CW^B + 2K \cdot CW^A = O(W^B)$ and $|T - T| \leq 2KC^2 W^{2A} = O(W^{A+B})$. Its exponent is $B/(A+B) = (\alpha_1 + \alpha_2)/(1 + \alpha_1 + \alpha_2)$.

Finally, given n, take the least W with $W^a \geq n$, where $a = a_\tau$ for a minimizing tree, and any n-element subset of F_W. Since $W \leq 2n^{1/a}$ and passing to a subset does not enlarge $\mathcal{D}$, we get $h_d(n) \leq C_\tau 2^a n^{u_d}$. Translating into $\mathbb{Z}^d_{\geq 0}$ gives integer sets by Lemma 2.1. □

Example 3.4. Gluing two intervals of length W with $K = W$ levels gives W^3 points with $|\mathcal{D}| = W^2 + 2(W-1)(2W-1) \sim 5W^2$, recovering the Freiman–Lev exponent $2/3$. Gluing an interval of length W^3 with this planar set, using $K = W^2$, gives $h_3(n) \ll n^{5/8}$.

4. Pruning lemmas

Throughout this section $E \subset \mathbb{R}^p \times \mathbb{R}^s$ is finite with $|E| = m$, $X = \pi_1 E \subset \mathbb{R}^p$ and $Y = \pi_2 E \subset \mathbb{R}^s$ are its two block projections, and for $y \in Y$ we write $E^y = \{x \in \mathbb{R}^p : (x, y) \in E\}$ for the horizontal fiber.

Lemma 4.1 (One-sided pruning)**.** *Let $a, b, c_p, c_s > 0$. Suppose $M(H) \geq c_p |H|^a$ for every nonempty $H \subseteq E^y$ $(y \in Y)$ and $M(J) \geq c_s |J|^b$ for every nonempty $J \subseteq Y$. Then*

$$M(E) \;\geq\; c_p c_s \Big(\frac{m}{2|Y|}\Big)^a \Big(\frac{m}{2|X|}\Big)^b.$$

Proof. Let $\lambda = c_p(m/(2|Y|))^a$ and $B^y = \{x \in E^y : |(E^y - x)^+| < \lambda\}$. For $x \in B^y$ we have $(B^y - x)^+ \subseteq (E^y - x)^+$, so $M(B^y) < \lambda$, and the hypothesis forces $|B^y| < m/(2|Y|)$. Hence fewer than $m/2$ points of E are *bad*, and some $x_0 \in X$ has a set $V = \{y : (x_0, y) \in E,\ x_0 \notin B^y\}$ with $|V| > m/(2|X|)$. Choose $y_0 \in V$ with $|(V - y_0)^+| = M(V) \geq c_s |V|^b$, and for each $w \in (V - y_0)^+$ choose $y_w \in V$ with $(y_w - y_0)^+ = w$. For $x \in E^{y_w}$,

$$\big((x, y_w) - (x_0, y_0)\big)^+ = \big((x - x_0)^+,\, w\big),$$

and $(x - x_0)^+$ takes at least λ values because $x_0 \notin B^{y_w}$. The sets obtained for different w are disjoint, and $(x_0, y_0) \in E$. Hence $|(E - (x_0, y_0))^+| \geq \lambda\, M(V)$. □

5. Two corners and constant factors in every dimension

For $a, b \in \mathbb{R}^d$ put $\operatorname{box}(a, b) = \prod_i [\min(a_i, b_i), \max(a_i, b_i)]$.

Lemma 5.1 (Two-corner selection)**.** *For $d \geq 1$, every nonempty finite $S \subset \mathbb{R}^d$ has $a, b \in S$, possibly equal, such that*

$$|S \cap \operatorname{box}(a, b)| \geq 2^{-(2^d+2)} |S|.$$

Proof. The selection principle is classical; see [ABKKW, Section 7]. For an explicit constant put $r = 2^{2^{d-1}}$. Iterated Erdős–Szekeres [ES35], sorting the first coordinate and taking successive monotone subsequences in the others, puts one of every $r+1$ points between two others. Coordinate ties may be ranked consistently; monotone ranks imply weak monotonicity of the original values.

The hypergraph of such triples has independence number at most r. If $N = |S| \geq 2r$, retain vertices independently with probability $p = 2r/N$ and delete at most one vertex for each surviving edge. If there are e edges, this gives $r \geq Np - ep^3$, hence $e \geq N^3/(8r^2)$. Assign each triple to a witnessing pair of corners. Some pair has at least $e/\binom{N}{2} \geq N/(4r^2)$ distinct middle points in its box. If $N < 2r$, a singleton suffices. Finally $4r^2 = 2^{2^d+2}$. □

Theorem 5.2. *Every nonempty finite $S \subset \mathbb{R}^d$, $d \geq 1$, satisfies*

$$M(S) \geq 2^{-2^{d+1}} |S|^{u_d}.$$

Proof. Put $A_d = 2^{d+1}$, $L_d = 2^d + 2$, and induct on d. The scalar case follows by anchoring at the minimum. Choose $E = S \cap \mathrm{box}(a, b)$ as above, so $m = |E| \geq 2^{-L_d} N$. With $I = \{i : a_i \leq b_i\}$ and $J = I^c$, the two actual anchors give

$$(x-a)^+ = (x_I - a_I, 0_J), \qquad (x-b)^+ = (0_I, x_J - b_J) \quad (x \in E).$$

Consequently $P_I(E), P_J(E) \leq M(S) = M$, including ties. If a block is empty, $M \geq m \geq 2^{-A_d} N^{u_d}$. Otherwise let $p = |I|$, $s = |J|$, and $t = u_p + u_s$. Lemma 4.1 and the induction give

$$M^{1+t} \geq 2^{-A_p - A_s} (m/2)^t \geq 2^{-[A_p + A_s + t(L_d+1)]} N^t.$$

For every nonempty split, $A_p + A_s \leq A_d$ and $L_d + 1 \leq A_d$. Taking the $(1+t)$-th root and using $t/(1+t) \geq u_d$ proves the claim. Together with Theorem 3.3, this completes the constant-factor assertion in Theorem 1.2. □

Remark 5.3. The loss in Lemma 5.1 cannot be replaced universally by $e^{-O(d)}$. List the $m = 2^{d-1}$ sign vectors $\sigma_j \in \{-1, 1\}^d$ having first entry 1, and form $v(\eta) = \sum_{j=0}^{m-1} 3^j \eta_j \sigma_j$ for $\eta \in \{0, 1\}^m$. The leading differing digit determines every coordinate comparison. In three points ordered by their first coordinates, the two successive leading digits differ, so their sign vectors disagree in some coordinate. No point is between the other two. Thus each two-corner box contains only its two corners, out of $2^{2^{d-1}}$ points. Each coordinate is nevertheless injective, so $M(S) = |S|$. This obstructs stronger box selection, not a theorem allowing a large-fan alternative.

6. Fiber-regular sets: aggregate pruning

Lemma 6.1. *For every finite $S \subset \mathbb{R}^d$ and $J \subseteq [d]$, $P_J(S)\Delta_J(S) \leq |S - S|$. Hence $R(S) \leq |S - S|/|S|$.*

Proof. Let H be a largest J-fiber, and choose $s_z \in S$ with $\pi_J s_z = z$ for every $z \in \pi_J S$. The sets $s_z - H \subseteq S - S$ have $|H|$ elements each and are pairwise disjoint, since their J-projections are distinct. □

Throughout this section $S \subset \mathbb{R}^d$ is fixed, $N = |S|$, $P_J = P_J(S)$, $\Delta_J = \Delta_J(S)$ and $R = R(S)$; all projection and fiber sizes refer to S. For a nonempty $A \subseteq S$ and $I \subseteq [d]$ we put

$$M_I(A) = \max_{a \in A} \big|\{(x-a)^+ : \ x \in A, \ \pi_I x = \pi_I a\}\big|,$$

the largest fan of A taken inside a single I-fiber. Thus $M_\varnothing(A) = M(A)$. Note that A may meet many I-fibers.

Lemma 6.2 (Relative projection balance)**.** *Let $A \subseteq S$ be nonempty, $I \subseteq [d]$, $m = d - |I| \geq 1$, $\rho = |A|/N$ and $M_I(A) \leq M$. Then the coordinates $[d] \setminus I$ can be partitioned as $U \sqcup V$ (one part possibly empty) so that*

$$\frac{P_{I\cup U}}{P_I} \leq Q \quad \text{and} \quad \frac{P_{I\cup V}}{P_I} \leq Q, \qquad Q = 2^{m+2}\frac{R}{\rho}M.$$

Proof. Let $n = |A|$. The number E of ordered pairs $(x, y) \in A^2$ with $\pi_I x = \pi_I y$ satisfies $E \geq n^2/P_I$, as A meets at most P_I fibers of π_I. For such a pair let $J(x, y) = \{j \notin I : x_j \geq y_j\}$. Fix $y \in A$ and $J \subseteq [d] \setminus I$. For the points x in the I-fiber of y with $J(x, y) = J$, the vector $(x - y)^+$ determines $\pi_{I\cup J}x$; it takes at most M values, and each value of $\pi_{I\cup J}x$ is shared by at most $\Delta_{I\cup J} \leq RN/P_{I\cup J}$ points. If $P_{I\cup J}/P_I > Q$, there are therefore fewer than $MRN\rho/(2^{m+2}RMP_I) = n/(2^{m+2}P_I)$ such x. Summing over the 2^m masks and the n anchors, fewer than $E/4$ pairs have $P_{I\cup J(x,y)}/P_I > Q$, and likewise fewer than $E/4$ have $P_{I\cup J(y,x)}/P_I > Q$. Take a pair avoiding both, let $U = J(x, y)$ and $V = ([d] \setminus I) \setminus U$. Then $V \subseteq J(y, x)$, and $P_{I\cup V} \leq P_{I\cup J(y,x)}$ by monotonicity of projection sizes. □

Lemma 6.3. *Let $m \geq 2$, $p + s = m$ with $p, s \geq 1$, and $t = u_p + u_s$. Then $\dfrac{\alpha_p + \alpha_s + t}{1 + t} \leq \alpha_m$.*

Proof. Recall $1 \leq t \leq 2$ and $\alpha_m \geq 1$ for $m \geq 2$. If $p, s \geq 2$, then with $L = \log_2 m$ we have $\alpha_p + \alpha_s = 1 + \frac{1}{2}\log_2(ps) \leq L$, and $(L + t)/(1 + t) \leq (L + 1)/2$ is equivalent to $(L - 1)(t - 1) \geq 0$. If $p = 1$, then $\alpha_p + \alpha_s = \alpha_{m-1} \leq \alpha_m$, and $(\alpha_m + t)/(1 + t) \leq \alpha_m$ because $\alpha_m \geq 1$. □

Proof of Theorem 1.3. We prove by induction on m the following statement: *if $I \subseteq [d]$, $m = d - |I| \geq 1$, and $A \subseteq S$ is nonempty, then with $\rho = |A|/N$ and $z = |A|/P_I$,*

$$M_I(A) \;\geq\; 256^{-m} R^{-\alpha_m} \rho^{\alpha_m} z^{u_m}. \tag{4}$$

Taking $I = \varnothing$ and $A = S$ (so $\rho = 1$, $z = N$) gives the theorem.

If $z \leq 1$, the right-hand side of (4) is at most $1 \leq M_I(A)$. If $m = 1$, some I-fiber of A contains at least z points, and anchoring at the one with smallest free coordinate gives at least z outputs. So let $m \geq 2$ and $z > 1$, write $n = |A|$, $M = M_I(A)$, and take U, V from Lemma 6.2.

If one part is empty, the other is $[d] \setminus I$, so $N/P_I \leq Q$, i.e. $M \geq z/(2^{m+2}R)$; this implies (4) because $\alpha_m \geq 1$, $R \geq 1$, $\rho \leq 1$, $z > 1$ and $u_m \leq 1$.

Otherwise let $p = |U|$, $s = |V|$ and $t = u_p + u_s$, and put

$$\lambda = 256^{-p} R^{-\alpha_p} \Big(\frac{\rho}{2}\Big)^{\alpha_p} \Big(\frac{n}{2P_{I\cup V}}\Big)^{u_p}, \qquad \lambda' = 256^{-s} R^{-\alpha_s} \Big(\frac{\rho}{2}\Big)^{\alpha_s} \Big(\frac{n}{2P_{I\cup U}}\Big)^{u_s}.$$

Call $a \in A$ *bad* if its fan inside its $(I \cup V)$-fiber of A has fewer than λ elements, and let B be the set of *all* bad points, across all such fibers. If $|B| \geq n/2$, then (4) for B with conditioning set $I \cup V$ (density at least $\rho/2$ and $|B|/P_{I\cup V} \geq n/(2P_{I\cup V})$) gives a point of B whose fan inside its $(I \cup V)$-fiber of B, hence of A, has at least λ elements, a contradiction. So $A' = A \setminus B$ has more than $n/2$ points.

By (4) for A' with conditioning set $I \cup U$, there is $a_0 \in A'$ such that $W = \{(y - a_0)^+ : y \in A',\ \pi_{I\cup U}y = \pi_{I\cup U}a_0\}$ has at least λ' elements. For each $w \in W$ choose such a y_w with $(y_w - a_0)^+ = w$. As y_w is not bad, the vectors $(x - y_w)^+$ with $x \in A$ and $\pi_{I\cup V}x = \pi_{I\cup V}y_w$ take at least λ values. For such x, since y_w agrees with a_0 on $I \cup U$ and x agrees with y_w on $I \cup V$,

$$(x - a_0)^+ = \big(0 \text{ on } I,\ \ (x - y_w)^+ \text{ on } U,\ \ w \text{ on } V\big),$$

and x lies in the I-fiber of a_0. Distinct w have distinct V-parts. Hence $M \geq \lambda\lambda'$.

By Lemma 6.2, $n/(2P_{I\cup V}) \geq z/(2Q) = z\rho/(2^{m+3}RM)$, and similarly for U. Substituting,

$$M^{1+t} \;\geq\; 2^{-8m-(\alpha_p+\alpha_s)-(m+3)t}\, R^{-(\alpha_p+\alpha_s+t)}\, \rho^{\alpha_p+\alpha_s+t}\, z^t.$$

By Lemma 6.3, $t/(1+t) \geq u_m$, and $8m+(\alpha_p+\alpha_s)+(m+3)t \leq 8m(1+t)$ (as $\alpha_p+\alpha_s \leq 2m \leq 2mt$ and $t \geq 1$), taking $(1+t)$-th roots and using $R \geq 1$, $\rho \leq 1$, $z > 1$ proves (4). □

Proof of Corollary 1.4. Let $q = |\mathcal{D}(F)|$ and $n = |F|$. By Lemmas 2.2 and 6.1, $R(F) \leq |F - F|/n \leq q^2/n$, and $q \geq M(F)$. Theorem 1.3 gives $q^{1+2\alpha_d} \geq 256^{-d} n^{u_d+\alpha_d}$, and $(u_d+\alpha_d)/(1+2\alpha_d) = \frac{1}{2} + (u_d - \frac{1}{2})/B_d$. Finally use Lemma 3.2. □

7. The unrestricted problem

Theorem 7.1 (Elementary original-coordinate extraction)**.** *Let $S \subset \mathbb{R}^D$ be finite, $N = |S|$, and $|S - S| \leq KN$. There is a subset $T \subseteq S$, of size at least $N/(1024K^{10})$, whose coordinate-prefix tree in any prescribed order has exactly b_i children at each retained vertex of level $i-1$. At most $\lceil \log_2(2K^2) \rceil$ of the b_i exceed one. Projection onto those original coordinates is injective on T.*

Proof. Compress coordinate i by replacing every nonempty line parallel to that coordinate by $\{0, \dots, m-1\}$, where m is its cardinality, keeping other coordinates fixed. This does not increase the sumset: over each fixed sum of the other coordinates, the compressed sum fiber is an initial interval of size

$$\max_{y+y'=z} (|S_y| + |S_{y'}| - 1),$$

whereas the original fiber contains $S_y + S_{y'}$ for every such pair, each of at least that size.

Compress in reverse order $D, D-1, \dots, 1$, obtaining a downset $A \subset \mathbb{Z}^D_{\geq 0}$ with $|A| = N$ and $|A + A| = \sigma N \leq K^2 N$ by (3). More precisely, A consists of $(b_1 - 1, \dots, b_D - 1)$ for all feasible exact branching profiles of the original prefix tree. Induct on depth to see this: the last $D-1$ compressions give the feasible suffix-profile downset separately inside each first-coordinate fiber. For a given suffix profile, the last compression records exactly the number of original fibers supporting it. Every point of A therefore lifts to an actual subtree with that branching profile and unchanged labels.

For every $a \in A$, the $2^{|\operatorname{supp} a|}$ vectors with coordinates $2a_i$ or $2a_i - 1$ when $a_i > 0$, and zero otherwise, belong to $A+A$. These cells are disjoint, since their coordinatewise rounded-up halves recover a. Thus

$$\sum_{a \in A} 2^{|\operatorname{supp} a|} \leq \sigma N.$$

Let $r = \lceil \log_2(2\sigma) \rceil$ and $E = \{a \in A : |\operatorname{supp} a| \leq r\}$, so $|E| \geq N/2$. Put $B = \max_{a \in E} \prod_i (a_i + 1)$. Every representation $z = a + b$ with $a, b \in E$ gives

$$1 \leq B \prod_i [(a_i + 1)(b_i + 1)]^{-1/2}.$$

The scalar sum $\sum_{x=0}^{z} [(x+1)(z-x+1)]^{-1/2}$ is one at $z = 0$ and at most four otherwise: split $j = x + 1$ at $(z+2)/2$ and use $\sum_{j \leq t} j^{-1/2} \leq 2\sqrt{t}$. Since $z \in E + E$ has at most $2r$ nonzero coordinates, its representation count is at most $16^r B$. Hence

$$N^2/4 \leq |E|^2 \leq 16^r B |E + E| \leq 16^r B \sigma N.$$

As $16^r \leq 256\sigma^4$, we have $B \geq N/(1024\sigma^5) \geq N/(1024K^{10})$. Lift a maximizing corner of E to its feasible original subtree. Its branching levels are the nonzero coordinates of that corner, at most $r \leq \lceil \log_2(2K^2) \rceil$. Two different retained leaves first differ at a branching level, proving injectivity of the stated projection. No assertion that compression preserves positive differences is used. □

Proof of Theorem 1.1. Let A be a set of n positive integers, $F = F_A \subset \mathbb{Z}^D$, and $q = |\mathcal{D}(F)| = |Q(A)|$. Put $K_0 = q^2/n \geq 1$ and $T = \log(2K_0)$. By Lemma 2.2(2), $|F - F| \leq K_0 n$. Theorem 7.1 supplies an actual subset and an injective original-coordinate projection with image $S \subset \mathbb{R}^d$ such that $m = |S| \geq n/(1024K_0^{10})$ and $d \leq \lceil \log_2(2K_0^2) \rceil \leq CT$. If $m = 1$, then $n \leq 1024K_0^{10}$, giving a stronger conclusion. Otherwise $d \geq 1$. Projection commutes with positive parts, so $q \geq |\mathcal{D}(S)|$, while $|S - S| \leq |F - F|$ and $R(S) \leq 1024K_0^{11}$. Theorem 1.3 and Lemma 3.2 give

$$\sqrt{K_0 n} = q \geq M(S) \geq 256^{-d}(1024K_0^{11})^{-\alpha_d} \left(\frac{n}{1024K_0^{10}}\right)^{\frac{1}{2}+\frac{1}{4d}}.$$

Taking logarithms, with $X = \log n$,

$$\frac{X}{4d} \;\leq\; C\big(d + T(1 + \log d)\big)$$

for an absolute constant C. Since $d \leq CT$, this gives $X \leq C'T^2(1 + \log(T + 1))$, hence $T \geq c\sqrt{X/\log X}$ for large X (if $T \geq X$ there is nothing to prove). Finally $q = \sqrt{n}\exp((T - \log 2)/2)$. □

8. Structured sets

Theorem 8.1 (Structured sets)**.** *Let $d \geq 1$.*

1. *Let $\Gamma \subset \mathbb{R}^d$ be a free abelian group of rank r, and let $S \subset \Gamma$ be finite, nonempty and midpoint-closed: $(x + y)/2 \in S$ whenever $x, y \in S$ and $x + y \in 2\Gamma$. Then $M(S) \geq 2^{-d-3r}|S|^{u_d}$. In particular every finite nonempty set $S = C \cap \Lambda$, with $C \subset \mathbb{R}^d$ convex and $\Lambda \subset \mathbb{R}^d$ a lattice of rank d, satisfies $M(S) \geq 16^{-d}|S|^{u_d}$.*
2. *Every set obtained from finite subsets of $\mathbb{R}$ by repeatedly forming ordered layer products (Definition 8.3) satisfies $M(S) \geq 4^{-(d-1)}|S|^{u_d}$.*

Midpoint-closed sets. For $S \subset \mathbb{R}^d$ write $S^+ = (S - 0)^+ = \{x^+ : x \in S\}$.

Proposition 8.2. *Let $\Gamma \subset \mathbb{R}^d$ be a free abelian group of rank r, and let $S \subset \Gamma$ be finite, midpoint-closed, symmetric ($S = -S$) and contain 0. Then $|S^+| \geq 2^{-d-2r}|S|^{u_d}$.*

Proof. Induction on d; for $d = 1$, $|S^+| \geq (|S| + 1)/2$. Let $n = |S|$ and $q = |S^+|$. Assign each $x \in S$ the set $\sigma(x) = \{i : x_i \geq 0\}$; some class E has at least $n/2^d$ points. If $\sigma = [d]$ or $\sigma = \varnothing$, then $q \geq |E|$ (using $S = -S$ in the second case). Otherwise let $U = \sigma$ and $V = [d]\setminus U$. For $x \in E$ we have $x^+ = (\pi_U x, 0)$ and $(-x)^+ = (0, -\pi_V x)$, and both lie in S^+; hence $|\pi_U E|, |\pi_V E| \leq q$.

Let $\Gamma_U = \Gamma \cap \ker \pi_V$ and $\Gamma_V = \Gamma \cap \ker \pi_U$, of ranks r_U, r_V. These subgroups have disjoint coordinate supports, so $r_U + r_V \leq r$. A largest V-fiber of E has at least $n/(2^d q)$ points, all in one coset of Γ_U; choose a class modulo $4\Gamma_U$ containing at least $n/(2^d q 4^{r_U})$ of them, and fix z in it. For every x in this class, $x + (-z) \in 4\Gamma_U \subseteq 2\Gamma$ and $-z \in S$, so $(x - z)/2 \in S \cap 2\Gamma_U =: S_U$. Thus $|S_U| \geq n/(2^d q 4^{r_U})$, and similarly $|S_V| \geq n/(2^d q 4^{r_V})$ for $S_V = S \cap 2\Gamma_V$. The set S_U is symmetric, contains 0, is midpoint-closed with respect to the rank-r_U group $2\Gamma_U$, and lives in the coordinates U; similarly for S_V. If $x \in S_U$ and $y \in S_V$, then $x + y \in 2\Gamma$, so $(x + y)/2 \in S$, and $((x + y)/2)^+$ equals $x^+/2$ on U and $y^+/2$ on V. Hence $q \geq |S_U^+|\,|S_V^+|$.

With $\alpha = u_{|U|}$, $\beta = u_{|V|}$ and $t = \alpha + \beta$, the induction hypothesis gives

$$q^{1+t} \;\geq\; 2^{-d(1+t)}\, 4^{-r_U(1+\alpha)-r_V(1+\beta)}\, n^t \;\geq\; 2^{-d(1+t)} 4^{-r(1+t)} n^t,$$

and $t/(1 + t) \geq u_d$ completes the induction. □

Proof of Theorem 8.1(1). Count pairs $(x, y) \in S^2$ with $x \equiv y \pmod{2\Gamma}$; there are at least $|S|^2/2^r$ of them, and each has its midpoint in S. Hence some $a \in S$ is the midpoint of at least $|S|/2^r$ such pairs, so $S' = S \cap (2a - S)$ has at least $|S|/2^r$ points. The set $S' - a$ is

symmetric, contains 0, lies in Γ and is midpoint-closed. By Proposition 8.2, $M(S) \geq |(S'-a)^+| \geq 2^{-d-2r}(|S|/2^r)^{u_d} \geq 2^{-d-3r}|S|^{u_d}$. A convex lattice set $C \cap \Lambda$ is midpoint-closed with respect to Λ, of rank d. □

Ordered layer products.

Definition 8.3. Let $A_1, \dots, A_k \subset \mathbb{R}^p$ and $B_1, \dots, B_k \subset \mathbb{R}^s$ be finite and nonempty, with every coordinate of every point of A_{i+1} larger than the corresponding coordinate of every point of A_i, and every coordinate of every point of B_{i+1} smaller than the corresponding coordinate of every point of B_i. Then $\bigcup_i (A_i \times B_i) \subset \mathbb{R}^{p+s}$ is an *ordered layer product* of the A_i and B_i. The sets of Proposition 3.1 are of this form.

Lemma 8.4. *For an ordered layer product S as above, with $a_i = |A_i|$ and $b_i = |B_i|$,*

$$M(S) \;\geq\; \max_i \Big(M(A_i)M(B_i) + \sum_{j>i} a_j + \sum_{j<i} b_j \Big).$$

Proof. Let $(x, y) \in A_i \times B_i$, with x and y realizing $M(A_i)$ and $M(B_i)$. Points of the same layer give the vectors $((x'-x)^+, (y'-y)^+)$, which take $M(A_i)M(B_i)$ values. A point (x', y') of a layer $j > i$ gives $(x'-x, 0)$, with $x'-x$ coordinatewise positive and larger than every vector $(x''-x)^+$, $x'' \in A_i$; these give a_j new values for each $j > i$. Symmetrically, layers $j < i$ give $(0, y'-y)$, with b_j new values each. All these values are distinct. □

Proof of Theorem 8.1(2). Induction on the construction, the case of subsets of $\mathbb{R}$ being trivial. Let S be an ordered layer product whose factors satisfy $M(A_i) \geq c_p a_i^{\alpha}$ and $M(B_i) \geq c_s b_i^{\beta}$ with $c_p = 4^{-(p-1)}$, $c_s = 4^{-(s-1)}$, $\alpha = u_p$, $\beta = u_s$. Put $q = M(S)$, $c = c_p c_s$, $t = \alpha + \beta$ and $\mu = \min(\alpha, \beta)$. By Lemma 8.4, $\sum_{i>1} a_i \leq q$, $\sum_{i<k} b_i \leq q$ and $a_i^{\alpha} b_i^{\beta} \leq q/c$ for every i. For $1 < i < k$ this gives $\min(a_i, b_i) \leq (q/c)^{1/t}$, hence $\sum_{1<i<k} a_i b_i \leq (q/c)^{1/t} \sum_{1<i<k}(a_i + b_i) \leq 2q(q/c)^{1/t}$; and the two end layers satisfy $a_i b_i \leq (q/c)^{1/\mu}$. Thus

$$|S| \leq 2q(q/c)^{1/t} + 2(q/c)^{1/\mu}.$$

If the second term is at least $|S|/2$, then $q \geq c(|S|/4)^{\mu} \geq 4^{-(p+s-1)}|S|^{u_{p+s}}$, as $\mu \geq u_{p+s}$. Otherwise $q^{1+1/t} \geq c^{1/t}|S|/4$, so $q \geq c^{1/(1+t)}(|S|/4)^{t/(1+t)} \geq 4^{-(p+s-1)}|S|^{u_{p+s}}$, as $c \leq 1$ and $t/(1+t) \geq u_{p+s}$. □

9. Weighted energy and projection-critical cores

Write $q(S) = |\mathcal{D}(S)|$. For a finitely supported nonnegative weight define

$$\mathcal{E}(f) = \sum_{a+b=c+d} f(a)f(b)f(c)f(d), \qquad \Lambda(S) = \max_{\substack{f \geq 0,\ \operatorname{supp} f \subseteq S \\ \sum f^2 = 1}} \mathcal{E}(f).$$

Set $\Lambda(\varnothing) = 0$; a zero-coordinate singleton has norm one. All logarithms in this section are natural. The following arguments are independent of any unproved projection-balance assertion.

Lemma 9.1 (Norm and optimizer facts)**.** *If $N = |S|$ and $K = |S - S|/N$, then $N/K \leq \Lambda(S) \leq N$. An exact maximizer f exists. Writing $\lambda = \Lambda(S)$ and $\mu = f^2$, one has $\mu_x \leq 1/\lambda$. The law*

$$Q(a, b, c, d) = \lambda^{-1} f(a)f(b)f(c)f(d)\, \mathbf{1}_{a+b=c+d}$$

has all four marginals μ, with $H(Q) = \log \lambda + 2H(\mu)$. Moreover

$$\Lambda(A) \geq \lambda \mu(A)^2 \quad (\varnothing \neq A \subseteq S).$$

Proof. Constant weights and Cauchy–Schwarz on difference representations give the lower bound. Cauchy–Schwarz in each sum fiber gives $\mathcal{E}(f) \le N(\sum f^2)^2$. Compactness gives a maximizer. Put $R_x = \sum_{x+b=c+d} f(b)f(c)f(d)$. The first-order equations give $R_x = \lambda f(x)$ on positive coordinates; at a zero coordinate the same equality follows because a positive derivative would improve the normalized weight. Since $R_x \le \|f\|_2 \|f * f\|_2 = \sqrt{\lambda}$, the atom bound follows. The equations also give the four marginals, and direct evaluation of $-\mathbb{E}_Q \log Q$ gives the entropy identity.

Mixed-energy Hölder bounds the additive sum of four weights by the product of their energy fourth roots. Indeed, embed their supports in the finitely generated torsion-free group they span, identify it with $\mathbb{Z}^r$, and apply Fourier Hölder on its compact dual. The same representation gives the triangle inequality for $\mathcal{E}^{1/4}$. Applying mixed Hölder to f, f, f, g gives $\mathcal{E}(g) \ge \lambda \langle f, g\rangle^4$. Use $g = f\mathbf{1}_A/\sqrt{\mu(A)}$ when $\mu(A) > 0$; otherwise the claimed bound is trivial. □

Theorem 9.2 (Full projection-critical normalization)**.** *Let $S \subset \mathbb{R}^D$, $N = |S|$, $q_0 = q(S)$, and $K_0 = q_0^2/N$. Fix $b > 0$ and minimize $J_b(T) = q(T)^{2b}/\Lambda(T)^{b+1}$ over all nonempty projected subsets $T = \pi_I A$ of S, including the empty-coordinate singleton. Break ties by minimum cardinality, then minimum number of coordinates. Put $\lambda = \Lambda(T)$, $q = q(T)$, and $\alpha = (b+1)/(2b)$. Then*

$$\lambda \ge N K_0^{-(2b+1)}, \qquad q^2/\lambda \le K_0^{2+1/b}, \tag{5}$$

$$q(\pi_I A) \ge q \left(\frac{\Lambda(\pi_I A)}{\lambda} \right)^{\alpha} \quad (\varnothing \ne A \subseteq T), \tag{6}$$

$$\Lambda(\pi_I A) \le \lambda, \qquad q^2 \le \lambda^{1+1/b}. \tag{7}$$

Every exact maximizing law on T has full support, and no proper coordinate projection is injective. If T has $d > 0$ coordinates, it has an axis-parallel pair in every coordinate and full affine dimension d. Its maximizing law satisfies

$$q(A) \ge q\mu(A)^{1+1/b}, \qquad 0 \le H(\mu) - \log\lambda \le (2 + 1/b)\log K_0.$$

Proof. The candidate family is closed under taking subsets and coordinate projections, by lifting one representative per projected point. Put $\lambda_0 = \Lambda(S)$ and $K_\lambda = q_0^2/\lambda_0 \le K_0^2$, using $\lambda_0 \ge N/K_0$. Since $|B| \le q(B)^2$ and $\Lambda(B) \le |B|$, one has $J_b(B) \ge 1/\Lambda(B)$. Comparison with S gives $\lambda \ge \lambda_0 K_\lambda^{-b}$, proving the first bound. Minimality also gives

$$q^2/\lambda \le K_\lambda(\lambda/\lambda_0)^{1/b} \le K_0^{2+1/b},$$

because every candidate has at most N points. Comparison with every projected subset gives (6). Its palette is at most q, implying no norm inflation, and the singleton competitor gives the second part of (7).

A proper positive support of a maximizer has the same norm and no larger palette, contradicting the cardinality tie rule. An injective proper projection preserves all original additive quadruples, so its norm is at least λ, hence equal by the preceding bound. Its palette must then equal q, contradicting the coordinate tie rule. Deleting each coordinate must therefore identify a pair differing only there; their differences span all coordinate axes. The mass inequality follows from Lemma 9.1. Finally $\log\lambda \le H(\mu) \le \log|T| \le 2\log q$, proving the entropy-defect bounds. □

Proposition 9.3 (A sufficient quadratic core bound)**.** *Suppose there is an absolute C such that*

$$\log\Lambda(T) \le CD^2$$

for every core selected by Theorem 9.2 from a D-coordinate parent with $b = 4D$. Then Conjecture 1.5 holds.

Proof. First use Theorem 7.1. If the retained set is a singleton the conclusion is already stronger. Otherwise its dimension D is $O(1+\log K_0)$, its logarithmic size loss is $O(1+\log K_0)$, and its palette surplus is at most a fixed constant times K_0^{11}. Apply (5) to this set with $b = 4D$. The assumed bound then yields $\log N = O((1+\log K_0)^2)$ for the original cardinality. The known upper bound supplies the other half of Conjecture 1.5. □

Completion and exchanges. The preceding normalization only compares projected subsets of the input. Allowing every set with a compatible positive-difference palette gives a stronger extremality condition. This enlargement preserves the quantitative reduction.

Proposition 9.4 (Completion normalization)**.** *Let $Q_0 = \mathcal{D}(S)$ and retain the notation of Theorem 9.2. Minimize J_b over all coordinate masks I and all finite sets $A \subset \mathbb{R}^I$ satisfying $0 \in A$ and $\mathcal{D}(A) \subseteq \pi_I Q_0$, with the same tie rules. A minimizer T exists. It has all the projected-subset criticality properties above and satisfies the sharper estimates*

$$\Lambda(T) \geq N K_0^{-(2b+1)}, \qquad \frac{q(T)^2}{\Lambda(T)} \leq K_0^2.$$

In particular, with $\lambda = \Lambda(T)$, $q = q(T)$, $W = |T - T|$, and

$$L = \log \lambda, \qquad \Delta = 2\log q - L, \qquad \kappa = \log(W/\lambda),$$

one has $0 \leq \kappa \leq \Delta \leq \min(L/b, 2\log K_0)$.

Proof. Every admissible A lies in $\pi_I Q_0 - \pi_I Q_0$, because $a = a^+ - (-a)^+$ and $0 \in A$. There are therefore finitely many candidates. A translate of every projected subset of S is admissible, and the family is closed under translated projected subsets. The proof of Theorem 9.2 gives criticality, full support, the tie-rule consequences, and the norm-retention estimate.

Put $\lambda_0 = \Lambda(S)$ and $K_\lambda = q_0^2/\lambda_0 \leq K_0^2$. Minimality gives

$$\frac{q^2}{\lambda} \leq K_\lambda (\lambda/\lambda_0)^{1/b}.$$

If $\lambda \leq \lambda_0$, this is at most K_λ; if $\lambda \geq \lambda_0$, use $q \leq q_0$ instead. Finally $\lambda \leq |T| \leq W \leq q^2$, and comparison with the singleton gives $q^2 \leq \lambda^{1+1/b}$. These prove the stated inequalities. The completed set need not be a subset of S. □

Proposition 9.5 (Exchange inequality)**.** *Let T be a completed core, let f be any exact optimizer, and put $\mu = f^2$, $s = 1 + 1/b$. Extend f by zero and define*

$$R_f(x) = \sum_{a+b-c=x} f(a)f(b)f(c).$$

For $C \subseteq T$ and a finite X disjoint from T, set $A = (T \setminus C) \cup X$ and $Z_X = \lambda^{-2}\sum_{x\in X} R_f(x)^2$. If $A \neq \varnothing$ and $\mathcal{D}(A) \subseteq \mathcal{D}(T)$, then, with $r = q - q(A)$,

$$\mu(C) \geq Z_X + 1 - (1 - r/q)^{1/s} \geq Z_X + \frac{r}{sq}. \tag{8}$$

Conversely, $q(A)/q < (1 - \mu(C) + Z_X)^s$ certifies $J_b(A) < J_b(T)$.

Proof. Mixed-energy Hölder gives $\langle R_f, g\rangle \leq \lambda^{3/4}\mathcal{E}(g)^{1/4}$. Choose g to be the normalized restriction of R_f to A. Since $R_f = \lambda f$ on T,

$$\Lambda(A) \geq \lambda(1 - \mu(C) + Z_X)^2.$$

If the restriction vanishes, this inequality is trivial. Translate A by one of its points to obtain an admissible competitor; this preserves its palette and energy. Completion minimality gives $q(A)/q \geq (\Lambda(A)/\lambda)^{s/2}$, proving the first inequality in (8); concavity of $t^{1/s}$ gives the second. The strict version follows by substituting the same norm bound into $J_b(A)/J_b(T)$. □

This yields a useful sufficient covering condition. Let C_j be the lift of $c_j > 0$ points of $\pi_{I_j} T$, and choose X_j as in Proposition 9.5. Write $r_j = q - q((T \setminus C_j) \cup X_j)$ and $\mathcal{R}_j = r_j + sqZ_{X_j} > 0$. The largest I_j-fiber has size at most $W/P_{I_j}(T)$, and each optimizer atom has mass at most $1/\lambda$. Thus

$$P_{I_j}(T) \le \frac{sqe^{\kappa} c_j}{\mathcal{R}_j}. \tag{9}$$

Suppose nonnegative weights a_j fractionally cover the coordinates, meaning $\sum_{j:i\in I_j} a_j \ge 1$ for every i. If $\mathcal{R}_j/c_j \ge q^{\theta_j}/C'_j$ with $0 \le \theta_j \le 1$ and $C'_j \ge 1$, put

$$\rho = \sum_j a_j, \qquad \beta = \sum_j a_j(1-\theta_j), \qquad B = \sum_j a_j \log C'_j.$$

Fractional Shearer, applied to the same optimizer law throughout, and (9) imply

$$(1-\beta/2)L \le \rho(\kappa + \log s) + \beta\Delta/2 + B. \tag{10}$$

Indeed, $L \le H(\mu) \le \sum_j a_j H(\mu_{I_j}) \le \sum_j a_j \log P_{I_j}(T)$; substitute (9) and $\log q = (L+\Delta)/2$. Fractional Shearer follows by writing each projection entropy as a sum of conditional coordinate entropies and observing that conditioning on fewer preceding coordinates increases each term.

After elementary extraction, let d be the retained dimension and K_0 the extracted parent's palette surplus. Choose $D = \max\{d, \lceil 1 + \log K_0 \rceil\}$, pad with constant coordinates, and take $b = 4D$. The extraction bounds imply that D is $O(1 + \log K)$ in terms of the original palette surplus K. A uniform supply of these certificates with $\beta \le 2 - 1/D$, $\rho = O(1)$ and $B = O(D)$ would give $L = O(D^2)$: use $\Delta \le L/(4D)$ in (10), leaving at least $L/(4D)$ on the left, and $\kappa \le 2 \log K_0 = O(D)$ on the right. Proposition 9.3 has the same proof for completed cores, so this would establish Conjecture 1.5.

The existence of such certificates for every sufficiently large completed core is unproved. The exchange inequality evaluates an actual replacement; it does not produce enough replacements to cover the coordinates. In particular, child criticality cannot be inferred from root criticality without a loss. For a projected descendant B, that loss is

$$\delta(B) = \log q(B) - \frac{s}{2} \log \Lambda(B) - \left(\log q - \frac{s}{2} L \right) \ge 0.$$

Root comparison gives a factor $e^{-\delta(B)}$ in the corresponding child palette bound. Signed telescoping does not bound positive excursions or the sum over branches.

Other verified progress and limits. The accompanying Lean development proves weighted pruning with one actual anchor, stationary fan and reference-law budgets, and compatible cuts whenever a proper exposed fractional cover of weight at most two uses at most four masks. The exposure and cover hypotheses are essential: no theorem supplies such a cover for every critical core. It also formalizes counterexamples to several stronger intermediate assertions, including a dimension-only budget for the average cost of passing to reflected parents. These are obstructions to particular proof methods, not counterexamples to Conjecture 1.5.

The completed-core approach incorporates the additional freedom of replacing points without enlarging the palette. Later exact finite checks and constructions test this approach, but do not close its universal covering step. Neither those computations nor further formalization of the proved lemmas supplies the missing geometric theorem.

10. Formal verification and reproducibility

The main results have an accompanying formalization [SA26] in Lean 4 [MU21] with mathlib [Mathlib20], using Lean `v4.29.0-rc8` and Mathlib revision `698d2b68b870`. The completed release audit covers 844 local and vendored modules and 12,917 project declarations. These counts include definitions and generated declarations; they are not counts of mathematical theorems. The transitive axiom audit permits only `propext`, `Classical.choice`, and `Quot.sound`.

The canonical import closure was rebuilt from source in an isolated extraction, reusing the pinned external dependency cache but no old project artifacts. Generated finite certificates were partitioned into smaller kernel-checked computations to bound memory, without changing their public theorem statements. Later five-mask drafts and preserved upstream reference copies are outside this verification claim.

The principal endpoints are as follows; namespaces are `Erdos539Geometry`, except for the explicitly named threshold theorem.

- Theorem 1.1, with $c = 1/16000$, is proved by
  `positiveCofactorMinimum_super_sqrt_explicit`
  `Erdos539.cofactorThreshold_super_sqrt_explicit`.
- Theorem 1.2 is proved by
  `fixedDimensionalMinimum_all_dimension_isTheta`
  `primeSupportedMinimum_all_dimension_isTheta`.
  The lower-bound endpoint is `all_dimension_fan_lower_bound`.
- Theorems 1.3, 7.1 and 8.1, the gluing identities, and Appendix A are covered by the same receipt. The unrestricted lower-bound dependency closure uses elementary extraction and has no PFR imports.
- Theorem 9.2 and the conditional implication in Proposition 9.3 are formalized. Propositions 9.4–9.5 and the lifted exchange criterion (10) have written proofs in this paper but are not included in that completed Lean receipt.

The arithmetic interface requires care. This paper minimizes over positive integers. Formal Conjectures allows zero in its definition of `cofactorThreshold`. The development proves the comparison between these definitions; it does not identify them. Both displayed headline endpoints have the stated quantifiers, with no universal cut or critical-core hypothesis assumed. Conversely, the conditional stronger-bound theorems retain those hypotheses explicitly.

The companion proof archive includes the theorem map, source hashes, audit output, and reproduction instructions. As of 19 October 2026, the DeepMind catalogue still marks the main problem open [FC26]. Formal verification checks the encoded statements and proofs; it does not establish novelty or substitute for independent review of the exposition and its mathematical interfaces.

Appendix A. Improved constants through dimension fifteen

We retain the notation $E \subset \mathbb{R}^p \times \mathbb{R}^s$, $m = |E|$, $X = \pi_1 E$, $Y = \pi_2 E$ and $E^y = \{x : (x, y) \in E\}$ from Section 4. Recall $T(F) = \max\{M(F), M(-F)\}$.

Lemma A.1 (Two-sided anchored pruning)**.** *Let $a, b, c_p, c_s > 0$. Suppose $M(H) \geq c_p|H|^a$ and $M(-H) \geq c_p|H|^a$ for every nonempty $H \subseteq E^y$, and $T(J) \geq c_s|J|^b$ for every nonempty $J \subseteq Y$. Then*

$$T(E) \;\geq\; c_p c_s \Big(\frac{m}{4|Y|}\Big)^a \Big(\frac{m}{2|X|}\Big)^b.$$

Proof. Let $\lambda = c_p(m/(4|Y|))^a$, and delete from each E^y the anchors whose fan in E^y in *either* direction has fewer than λ elements. As in Lemma 4.1 (applied to H and to $-H$) each of the

two bad sets in E^y has fewer than $m/(4|Y|)$ elements, so fewer than $m/2$ points are deleted. Choose x_0 and V as before, with $|V| > m/(2|X|)$, and choose $y_0 \in V$ and a sign realizing $T(V)$. If the sign is $+$, argue as in Lemma 4.1. If it is $-$, for each $w \in (y_0 - V)^+$ choose $y_w \in V$ with $(y_0 - y_w)^+ = w$; then $((x_0, y_0) - (x, y_w))^+ = ((x_0 - x)^+, w)$ for $x \in E^{y_w}$, and $(x_0 - x)^+$ takes at least λ values. Thus one of the two fans of E at (x_0, y_0) has at least $\lambda T(V)$ elements. □

Lemma A.2 (Two-sided pruning for $\mathcal{D}$)**.** *Let $a, b, c_p, c_s > 0$. Suppose $T(H) \geq c_p|H|^a$ for every nonempty $H \subseteq E^y$, and $|\mathcal{D}(J)| \geq c_s|J|^b$ for every nonempty $J \subseteq Y$. Then*

$$|\mathcal{D}(E)| \;\geq\; c_p c_s \Big(\frac{m}{2|Y|}\Big)^a \Big(\frac{m}{4|X|}\Big)^b.$$

Proof. Let $\lambda = c_p(m/(2|Y|))^a$, and call $x \in E^y$ bad if both of its fans in E^y have fewer than λ elements. The bad set B^y satisfies $T(B^y) < \lambda$, so $|B^y| < m/(2|Y|)$, and more than $m/2$ points are good. Label each good point (x, y) by a sign ϵ such that its ϵ-fan in E^y has at least λ elements, and keep the more frequent label, say ϵ; at least $m/4$ points remain. Some x_0 has a set V of size at least $m/(4|X|)$ of kept points (x_0, y). For $w \in \mathcal{D}(V)$ choose $y, y' \in V$ with $w = (y - y')^+$ if $\epsilon = +$, and with $w = (y' - y)^+$ if $\epsilon = -$. If $\epsilon = +$, the vectors $((x, y) - (x_0, y'))^+ = ((x - x_0)^+, w)$, $x \in E^y$, take at least λ values; if $\epsilon = -$, so do $((x_0, y') - (x, y))^+ = ((x_0 - x)^+, w)$. Different w give disjoint sets, so $|\mathcal{D}(E)| \geq \lambda|\mathcal{D}(V)|$. □

Lemma A.3 (Planar fans)**.** *Every $H \subset \mathbb{R}^2$ with $|H| = N$ satisfies $M(H) \geq (N/2)^{2/3}$.*

Proof. Let r, s be the numbers of distinct first and second coordinates. By Lemma 2.2(4), $M(H) \geq \max(r, s)$. Put $k = \lceil N/(2s) \rceil - 1$ and $l = \lceil N/(2r) \rceil - 1$. On each horizontal line remove the k points with largest first coordinate (all of them if there are fewer), obtaining H_1; then on each vertical line remove from H_1 the l points with largest second coordinate. Fewer than $N/2 + N/2$ points are removed, so some a survives. Then a's vertical line contains at least $l + 1$ points $b \in H_1$ with $b_2 \geq a_2$, and each such b has at least $k + 1$ points $x \in H$ on its horizontal line with $x_1 \geq b_1 = a_1$. These $(k + 1)(l + 1) \geq N^2/(4rs)$ points are distinct and satisfy $x \geq a$ coordinatewise, so $|(H - a)^+| \geq N^2/(4rs)$. Multiplying the three lower bounds gives $M(H)^3 \geq N^2/4$. □

This is the anchored form of the argument of [HLP08, Lemma 2.2].

Lemma A.4 (Three-dimensional fans)**.** *Every $F \subset \mathbb{R}^3$ with $|F| = n$ satisfies $M(F) \geq 8^{-3}n^{5/8}$.*

Proof. For distinct $x, y \in F$, each coordinate is nonnegative in $x - y$ or in $y - x$, so one of them has at least two nonnegative coordinates. Counting ordered pairs (diagonal pairs included), some anchor $a \in F$ has at least $(n + 1)/2$ points x such that $x - a$ has at least two nonnegative coordinates. Classify these by their set of nonnegative coordinates; one of the four possible classes, E, has $m \geq n/8$ points. If it is $\{1, 2, 3\}$, then $M(F) \geq |(F - a)^+| \geq m$. Otherwise it is a two-element set P, and $Q = [3] \setminus P$. For $x \in E$ the vector $(x - a)^+$ determines $\pi_P x$, so $|\pi_P E| \leq M(F)$; and $|\pi_Q E| \leq P_Q(F) \leq M(F)$ by Lemma 2.2(4). Apply Lemma 4.1 to $E \subset \mathbb{R}^P \times \mathbb{R}^Q$, using Lemma A.3 ($c = 2^{-2/3}$, exponent $2/3$) and $M(H) = |H|$ in dimension one. With $M = M(F) \geq M(E)$ this gives $M^{8/3} \geq 2^{-2/3}(m/2)^{5/3} \geq 2^{-22/3}n^{5/3}$, which implies the claim. □

Theorem A.5. *Every finite $F \subset \mathbb{R}^d$ with $|F| = n$ satisfies*

$$T(F) \geq 8^{-d}n^{u_d} \quad (1 \leq d \leq 7), \qquad |\mathcal{D}(F)| \geq 8^{-d}n^{u_d} \quad (1 \leq d \leq 15).$$

Proof. The one-sided bounds $M(F) \geq 8^{-p}n^{u_p}$ hold for $p \leq 3$: trivially for $p = 1$, by Lemma A.3 for $p = 2$ and by Lemma A.4 for $p = 3$. They give the claim for T when $d \leq 3$.

Let $4 \le d \le 7$ and assume the bound for T in smaller dimensions. Translate so that $0 \in F$, and classify $x \in F$ by $\sigma(x) = \{i : x_i \ge 0\}$; some class E has $m \ge n/2^d$ points. If $\sigma \in \{\varnothing, [d]\}$, the fan of F at 0 in the appropriate direction is injective on E, so $T(F) \ge m \ge 8^{-d} n^{u_d}$. Otherwise write $[d] = P \sqcup N$ with $P = \sigma$. Since $(x-0)^+ = (\pi_P x, 0)$ and $(0-x)^+ = (0, -\pi_N x)$ for $x \in E$, both block projections of E have at most $T(F)$ elements. Let the smaller block, of dimension $p \le 3$, be the first block in Lemma A.1, with the one-sided bounds (which also apply to $-H$), and use the inductive bound for T on the other block. With $t = u_p + u_{d-p}$ we get

$$T(F)^{1+t} \;\ge\; 8^{-d}\,(m/4)^t \;\ge\; 8^{-d}\,2^{-(d+2)t}\,n^t.$$

Since $t/(1+t) \ge u_d$ and $3d + (d+2)t \le 3d(1+t)$, this gives $T(F) \ge 8^{-d} n^{u_d}$.

For $\mathcal{D}$ argue identically with $q = |\mathcal{D}(F)|$: the block projections of E have at most q elements since $(F-0)^+, (0-F)^+ \subseteq \mathcal{D}(F)$. For $d \le 15$ the smaller block has dimension $p \le 7$, so Lemma A.2 applies with the bound for T on that block and the inductive bound for $\mathcal{D}$ on the other, and the same arithmetic concludes. □

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